

Unit 9, Lesson 4: Adding and Subtracting Expressions with Rational Coefficients

Benchmark

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

Learning Target

- ☐ I can apply properties of operations to add and subtract linear expressions with rational coefficients.

9.4.1 Warm-Up: Bell Work

1. Rewrite each expression by combining like terms.

a. $5.7x - (3.5x + 6)$

b. $\left(-\frac{5}{6}y + 5\frac{1}{2}\right) - \left(\frac{2}{3}y + 34\right)$

9.4.2 Collaboration: Adding and Subtracting Linear Expressions

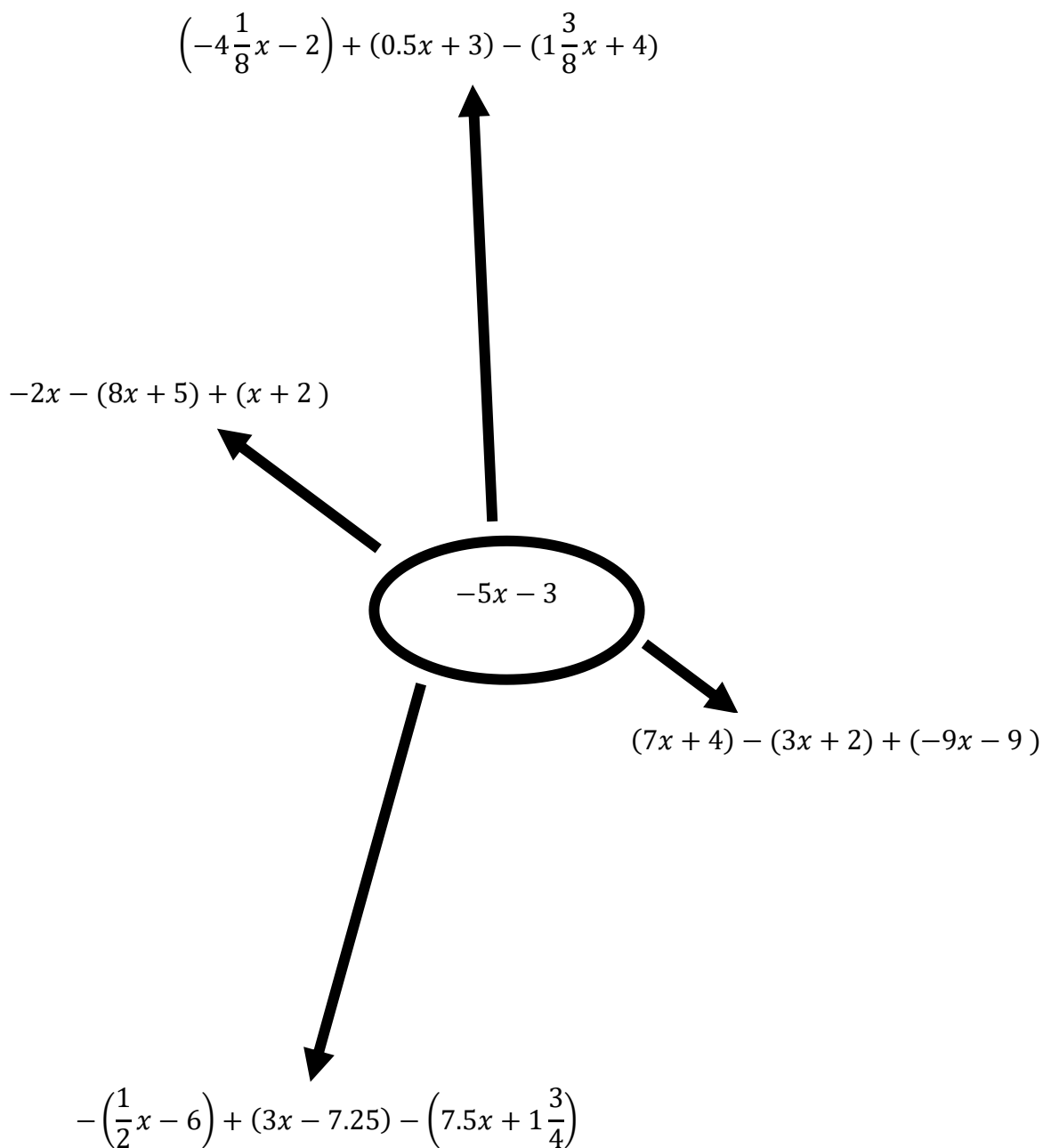
1. Two students made errors when rewriting the expression $(6x + 9) - (7 - x) + (8x + 11)$.

Zoe’s Work	Patricia’s Work
$(6x + 9) - (7 - x) + (8x + 11)$ $(6x + 9) - (7x + 18)$ $-x - 9$	$(6x + 9) - (7 - x) + (8x + 11)$ $(5x + 2) + (8x + 11)$ $13x + 13$

- a. Working with your partner, write an explanation to Zoe that describes her error.
- b. Write an explanation to Patricia that describes her error.
- c. Simplify the expression $(6x + 9) - (7 - x) + (8x + 11)$.

9.4.3 Guided Practice: Equivalent Expressions

1. Circle the expressions that are equivalent to $-5x - 3$ and cross out those that are not.



9.4.4 Your Turn!

1. Rewrite each expression by combining like terms.

a. $(1 - 4y) - (2.5y + 7) - (-0.3y + 3)$

b. $\left(3\frac{1}{2}x - 4\right) + (x - 3.2) - \left(-\frac{1}{4}x + 9\right)$

c. $-(2r + 5) - (6 - 3r) + (10r - 4)$

9.4.5 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

My favorite mistake from today was.

Mmeans
I
Sstart
To
Acquire
Knowledge
Experience
Skills

